



KAUVERY HOSPITAL: PROUD TO BE A LEADER IN ORGAN TRANSPLANTATION

HEART TRANSPLANTATION

Cardiovascular disease, is responsible for over 20 million deaths worldwide every year and remains the leading cause mortality in India and globally. Almost half of these deaths are from heart disease alone. Whilst acknowledging that the majority of morbidity and mortality from heart disease is preventable with small incremental changes in lifestyle and access to preventive health care, it is opportune, as we pass 2023 World Heart Day (WHD) on the 29th September, to reflect on transplantation of the human heart.

The first successful human heart transplant was carried out by Christiaan Barnard at the Groote Schuur Hospital in Capetown, South Africa in December 1967 with the recipient surviving only 18 days. It led to a flurry of similar procedures around the world including India. Just 6 weeks later, in February 1968, Dr PK Sen performed the fourth human heart transplant at the KEM Hospital in Mumbai. Sadly, the 27-year-old farmer only survived three hours. Early deaths after heart transplantation became the norm, and the tragic survival statistics led to a global moratorium on the procedure. What had been hailed as one of the most significant milestones in medicine, simply went back to the clinical drawing board. Considering the disappointing early results, perseverance by a small number of dedicated clinicians to change and effect acceptable outcomes

was truly remarkable. A combination of introducing modern immunosuppression drugs, novel technique for preservation of the heart permitting distant procurement of the donor organ, and refinement in surgical and post-operative management, yielded excellent outcomes. It is therefore noteworthy that in India, merely 10 months after Dr Barnard's seminal procedure, Dr P Venugopal at AIIMS Delhi, performed what became the second attempt but the country's first successful heart transplant with the recipient surviving another 14 years.

Around 9000 hearts are transplanted globally, with 75% of these being performed in the US and Europe. Because heart transplant necessarily must rely on utilising gifted donor hearts after death, the important metrics are the number of deceased organ donations and the number of heart transplants per million population (pmp). Data from the Global Observatory on Donation and Transplantation (www.transplant-observatory.org) shows that the rate of deceased organ donation is 6.86 pmp globally, 16.8 in Europe, 44.5 in the USA, and just 0.67 in India, which incidentally does almost 90% of the procedures in S E Asia. During 2022, at a rate of only 0.17 pmp, here were 243 heart transplants performed in India. However, a marginal increment in deceased

organ donation to 5 pmp, and a heart transplant rate equivalent to that of Europe at 3.5 pmp would result in almost 5000 procedures annually in India, and over 17000 if carried out at the rate of the USA at 12.4 pmp. The common conditions which cause end-stage heart failure are the following: Cardiomyopathy, a broad term encompassing disease of the heart muscle that leads to an enlarged heart with poor function; Ischaemic Heart Disease, a condition where the process of atherosclerosis leads to deposition of fatty material that narrow or block the arteries supplying blood to the

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